

Agentic AI: Foundations and Open-Source Practice

Day 1 Task

1. Task Details

Task title	Comparing a Plain Chatbot, a Rule-Based Workflow, and an AI Agent on a Scenario of Your Own
Unit	Unit 1: Foundations of AI Agents -- Agent = LLM + Tools + Loop
Task type	Individual conceptual analysis, based on a private-data scenario you choose yourself
Format required	A GitHub repository containing your project code, an output-screenshots folder, and a written analysis in Markdown
Submission	Link to your GitHub repository

2. Aim

Choose a small private-data scenario of your own and demonstrate how the same problem can be solved using a plain chatbot, a rule-based workflow, and a tool-using AI agent. Explain how each approach works, what data and tools it requires, how it handles the user's request, and where its limitations appear. Compare the three approaches based on flexibility, decision-making, tool usage, private-data access, multi-step task handling, automation, and reliability, and analyze which approach is most suitable for your chosen scenario and why. Your analysis should clearly demonstrate that a plain chatbot mainly provides responses using an LLM, a rule-based workflow follows predefined steps and conditions, while an AI agent combines an LLM + Tools + Loop to reason about the task, select and use appropriate tools, observe the results, and continue taking actions until the task is completed. Conclude by explaining when a chatbot, rule-based workflow, or AI agent would be the most appropriate choice for different types of problems.

3. Result

Your submission must produce the following, for the private-data scenario you chose:

3.1 Explanation of each approach

For each of the three approaches, explain:

- What data it uses, and whether it can access your private data at all.
- What tools, rules, or none of either it requires.
- How it handles a user's request from start to finish.
- Where its limitations show up on your specific scenario.

Your explanation must clearly show:

- A plain chatbot mainly provides responses using an LLM alone.
- A rule-based workflow follows predefined steps and conditions, with no LLM involved.
- An AI agent combines an LLM + Tools + Loop -- it reasons about the task, selects and uses appropriate tools, observes the results, and continues taking actions until the task is completed.

3.2 Comparison table

Fill in this table for your scenario, comparing the three approaches on each basis listed:

Basis for comparison	Plain chatbot	Rule-based workflow	AI agent
Flexibility			
Decision-making			
Tool usage			
Private-data access			
Multi-step task handling			
Automation			
Reliability			

3.3 Suitability analysis

State which of the three approaches is most suitable for your chosen scenario, and justify the choice using points from your comparison table above.

3.4 Conclusion

Close your analysis by explaining, in general, when a chatbot, a rule-based workflow, or an AI agent would be the most appropriate choice, for different types of problems -- not only for your own scenario.

4. Submission Guidelines

1. Push your complete project code (all files for the chatbot, workflow, and agent you built) to a GitHub repository.
2. Inside the repository, create a folder named Output and place the output screenshots of all three systems running on your scenario in it.
3. Inside the repository, create a file named analysis.md containing your full written analysis from Section 3 -- the explanation of each approach, the filled-in comparison table, the suitability analysis, and the conclusion -- written out in paragraphs, not just bullet points.
4. Submit the link to your GitHub repository.

Your analysis.md is what will be read and graded first, so make sure it stands on its own -- write it so that someone who has not seen your code can still follow your reasoning end to end.
